

Design and Analysis of Randomized Trials in Public Health
PUBH 526

Homework 12

You have designed an intervention intended to improve linear growth of infants in a resource-poor setting, and you will compare your intervention to standard of care in a randomized controlled trial using a completely randomized design. Infants (0-6 months old) will be assessed at baseline (prior to intervention), midline (during the intervention), and endline (after the intervention).

At each time point, you will measure their length (height measured while lying on a flat surface) and use that to calculate their height-for-age Z-score (HAZ). HAZ is a measure that compares a child's height to the average height of healthy children of the same age and sex, and the number is specifically the number of standard deviations a child is above or below the healthy age- and sex-standardized average. A HAZ of -1, for example, means a child is one standard deviation shorter than the average healthy child of their age and sex, and a HAZ < -2 leads to a child being classified as stunted, which indicates chronic undernutrition.

The dataset "growth" has HAZ data for 400 infants that were randomized to receive either the standard of care (treatment=1) or your new intervention (treatment=2). Each child ("ID") is measured at the three time points described above (time=1 for baseline, time=2 for midline, and time=3 for endline). Analyze these data using a longitudinal model (option 3 in Topic 14). What effect did the intervention have? Show your process, from exploring your data to formulating your hypotheses to thoroughly answering this question (including diagnostics to check assumptions made by your statistical model), and include your SAS code. (10 points)

Homework 12

Alexandra Chang

PUBH 526: Design and Analysis of Randomized Trials in Public Health

December 12, 2025

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Question 1 (10 points)

Analyze these data using a longitudinal model (option 3 in Topic 14). What effect did the intervention have? Show your process, from exploring your data to formulating your hypotheses to thoroughly answering this question (including diagnostics to check assumptions made by your statistical model), and include your SAS code.

Longitudinal Analysis of Infant Growth (HAZ)

This randomized controlled trial evaluated the effect of a nutritional intervention on infant linear growth in a resource-poor setting. Infants were randomized to receive either standard of care or the intervention and were measured repeatedly at baseline, midline, and endline. The outcome of interest was height-for-age Z-score (HAZ), a standardized measure of child linear growth. Because each infant contributed multiple observations over time, the repeated measurements within individuals are not independent and must be accounted for in the analysis.

To appropriately model the longitudinal structure of the data, a mixed-effects model was used following Option 3 from Topic 14. The model included fixed effects for treatment group, time, and the treatment-by-time interaction. Repeated measures were specified at the child level using a compound symmetry covariance structure, which assumes a constant correlation among measurements taken on the same child. This approach allows all available data to be used while accounting for within-child correlation.

The primary hypothesis of interest was whether the intervention affected growth trajectories over time. Specifically, the null hypothesis was that there was no treatment-by-time interaction, indicating that changes in HAZ over time were the same in both treatment groups. The alternative hypothesis was that the treatment-by-time interaction was nonzero, implying that infants receiving the intervention experienced a different pattern of linear growth over time compared to those receiving standard care. This interaction term represents the intervention effect and is equivalent to a difference-in-differences estimate.

Results from the mixed-effects model showed that the treatment-by-time interaction was **statistically significant**. This indicates that infants in the intervention group **did** experience a different trajectory of HAZ over the study period compared with infants in the control group. Because randomization ensures baseline comparability on average, this interaction can be interpreted as evidence for a causal effect of the intervention on growth over time.

Model diagnostics were assessed using residual plots generated by PROC MIXED. Residuals plotted against predicted values showed no strong patterns, suggesting that the assumption of constant variance was reasonable. Histograms and normal quantile plots of the residuals indicated approximate normality. Residual distributions were also similar across treatment groups and time points. Overall, there was no evidence of major violations of model assumptions.

In summary, a longitudinal mixed-effects model provided an appropriate framework for analyzing repeated HAZ measurements and estimating the intervention effect over time. This approach allowed for a direct assessment of whether the intervention altered infant growth trajectories while properly accounting for within-child correlation.

See SAS codes/outputs below:

```
1  /*****
2  * PUBH 526      Design and Analysis of Randomized Trials in Public
3  * Homework 12
4  * Topic: Repeated Measures / Longitudinal Analysis
5  * Dataset: growth.sas7bdat
6  * Author: Alexandra Chang
7  *****/
8
9  /* Reset environment */
10 options nocenter ls=80 pageno=1 nosyntaxcheck;
11 ods graphics on;
12 title;
13
14 /*****
15 * 1. Load Dataset
16 *****/
17
18 libname mydata '/home/u64258127/sasuser.v94';
19
20 data growth;
21     set mydata.growth;
22 run;
23
24 /*****
25 * 2. Explore Data Structure
26 *****/
27
28 proc contents data=growth;
29 run;
30
31 /*****
32 * 3. Descriptive Statistics by Treatment Group and Time
33 *****/
34
35 title "Mean HAZ by Treatment Group and Time";
36 proc means data=growth mean std;
37     class treatment time;
```

```

38     var haz;
39 run;
40
41 /*****
42  * 4. Longitudinal Mixed Model (Topic 14      Option 3)
43  *      Treatment      Time interaction estimates intervention effect
44  *      Diagnostic plots requested directly from PROC MIXED
45  *****/
46
47 title "Longitudinal Mixed Model with Treatment      Time Interaction";
48 proc mixed data=growth plots=all;
49     class id treatment time;
50     model haz = treatment time treatment*time / solution;
51     repeated time / subject=id type=cs;
52 run;
53
54 ods graphics off;

```

Listing 1: SAS Code for Homework 12

WORK GROWTH		Observations	1200
Data Set Name	DATA	Variables	4
Member Type	DATA	Indexes	0
Engine	V9	Observation Length	32
Created	12/12/2025 16:46:02	Deleted Observations	0
Last Modified	12/12/2025 16:46:02	Compressed	NO
Protection		Sorted	NO
Data Set Type			
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	4076
Obs in First Data Page	1200
Number of Data Set Repairs	0
Filename	/saswork/SAS_work4F8300010F1D_0daw01-usw2.sas.com/SAS_workC8500010F1D_0daw01-usw2.sas.com/growth.sas7bdat
Release Created	9.0401MB
Host Created	Linux
Index Number	530901613
Access Permission	rw-r--r--
Owner Name	u64258127
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes			
#	Variable	Type	Len
1	ID	Num	8
4	haz	Num	8
3	time	Num	8
2	treatment	Num	8

Mean HAZ by Treatment Group and Time

Analysis Variable : haz					
treatment	time	N	Obs	Mean	Std Dev
1	1	200	-0.7325500	0.4012919	
	2	200	-1.2418500	0.3658405	
	3	200	-1.3959000	0.3936917	
2	1	200	-0.6710500	0.3535184	
	2	200	-1.0005500	0.3883657	
	3	200	-1.1292500	0.3493655	

Longitudinal Mixed Model with Treatment * Time Interaction

Model Information	
Data Set	WORK.GROWTH
Dependent Variable	haz
Covariance Structure	Compound Symmetry
Subject Effect	ID
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information	
ID	Levels: 400
time	Levels: 3

Dimensions	
Covariance Parameters	2
Columns in X	12
Columns in Z	0
Subjects	400
Max Obs per Subject	3

Number of Observations	
Number of Observations Read	1200
Number of Observations Used	1200
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	1083.63188593	
1	1	527.79573624	0.00000000

Convergence criteria met.			
Cov Parameter	Subject	Estimate	
CS	ID	0.09545	
Residual		0.04584	

Fit Statistics			
-2 Res Log Likelihood	527.8		
AIC (Smaller is Better)	531.8		
AICC (Smaller is Better)	531.8		
BIC (Smaller is Better)	539.8		

Null Model Likelihood Ratio Test			
DF	Chi-Square	Pr > ChiSq	
1	555.84	<.0001	

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DF	Chi-Square	Pr > ChiSq	
1	555.84	<.0001	

Solution for Fixed Effects						
Effect	treatment	time	Estimate	Standard Error	DF	Pr > t
Intercept			-1.1293	0.02658	398	<.0001
treatment	1		-0.2666	0.03759	398	<.0001
treatment	2		0			
time		1	0.4592	0.02141	796	<.0001
time		2	0.1287	0.02141	796	<.0001
time		3	0			
treatment*time	1	1	0.2051	0.03028	796	<.0001
treatment*time	1	2	0.02535	0.03028	796	0.84
treatment*time	1	3	0			
treatment*time	2	1	0			
treatment*time	2	2	0			
treatment*time	2	3	0			

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
treatment	1	398	32.54	<.0001
time	2	796	742.28	<.0001
treatment*time	2	796	27.29	<.0001

